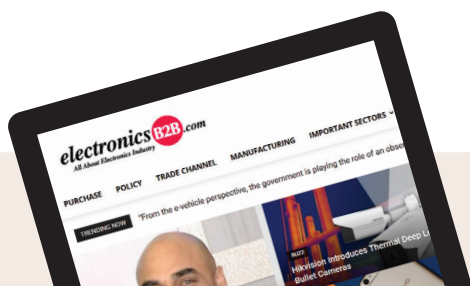


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Computer vision: Then and now

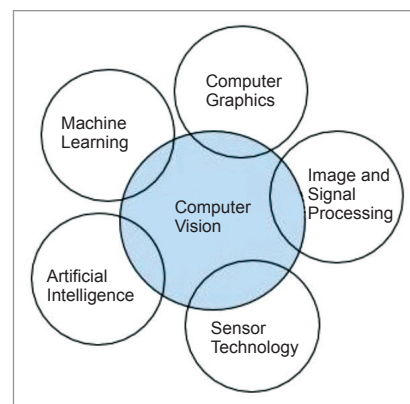
In his whitepaper titled 'Everything You Ever Wanted To Know About Computer Vision,' Ilija Mihajlovic, an iOS developer who is passionate about machine learning (ML), explains how ML and deep learning have simplified computer vision. Before deep learning came into the picture, computer vision did not have as much scope as it does today, mainly because a lot of manual coding was required, such as creating a database, annotating images, capturing new images, etc.

"With machine learning, developers no longer needed to manu-

ally code every single rule into their vision applications. Instead, they program features or smaller applications that could detect specific patterns in images," writes Mihajlovic. Today, we use deep learning for computer vision, and all of the work is done by neural networks. When we provide neural networks with lots and lots of labeled images of a particular object, they are able to identify patterns in them.

Hardware for computer vision

Capturing data is an important aspect of computer vision applications, and hardware platforms play a major role here. The main components required for computer vision



Fields that drive computer vision (Credit: IFA)

applications are cameras and image sensors for capturing images, I/Os, a communication interface, and a processing unit. Imaging sensors present in cameras are characterised by their resolution (number of pixels), speed (frames per second), and number of colours.

Depending on the type of computer vision application, the processing unit might vary from CPUs

Components of Computer Vision

Hardware: Cameras/Imaging sensors

Software: Machine learning models for processing the images

Hardware: Processing units or conditional logic to automate application-specific use cases